S6 Candidate-Specific Validation Update

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Contents

0.1 Result . . . 1
0.2 Status discipline . . . 2
0.3 Properness, surjectivity, and flatness . . . 2
0.3.1 3.1 Properness and surjectivity . . . 2
0.3.2 3.2 Flatness of f . . . 2
0.3.3 3.3 Flatness of vector bundles . . . 3
0.4 Top-degree cohomology and base change . . . 3
0.4.1 4.1 Vanishing of the next direct image . . . 3
0.4.2 4.2 The degree-two isomorphism . . . 3
0.5 Duality on the central fibre . . . 4
0.6 The conductor-supported differential . . . 4
0.6.1 6.1 Conormal injection . . . 4
0.6.2 6.2 Vanishing on the conductor and nonvanishing off it . . . 4
0.6.3 6.3 Normalization-conductor descent . . . 5
0.7 Triviality on the normalization . . . 5
0.8 Exact dependency boundary . . . 5
0.9 Preliminary audit of the candidate construction . . . 6
0.10 Candidate-specific test program . . . 6
0.10.1 CAND-X: existence of the global threefold . . . 6
0.10.2 CAND-W: exact central fibre . . . 6
0.10.3 TOP-H2: integral topology . . . 7
0.10.4 ANA-REF: publication-level references . . . 7
0.11 Conclusion . . . 7
0.12 Package contents . . . 7

Scope: properness, flatness, duality, the conductor-supported differential, and top-degree cohomology/base change in Sections 9-10 of the current 108-page candidate.
Disposition of this slice: verified from the stated candidate inputs.
Disposition of the whole candidate: not certified by this update.

0.1 Result

The new workbench observation is correct and materially narrows the search for a candidate-specific counterexample.

1. Once the constructed object is a compact connected complex threefold and the displayed map $f : X \to \mathbf{P}^1$ is a nonconstant holomorphic map, properness and flatness follow.
2. Every holomorphic vector bundle on X, in particular $E = T_X \otimes L$, is flat over $\mathbf{P}^1$.

3. For every $b \in \mathbf{P}^1$, including the nonreduced multiple fibres, the canonical top-degree base-change morphism
$$\beta_b^2 : (R^2 f_* E)_b \otimes_{\mathcal{O}_{\mathbf{P}^1,b}} k(b) \longrightarrow H^2(X_b, E|_{X_b})$$
is an isomorphism, with X_b the scheme-theoretic fibre.
4. The duality and local conductor argument at the reduced normal-crossings fibre W is sound. It gives
$$H^2(W, (T_X \otimes L)|_W) \neq 0$$
for every $L \in \mathrm{Pic}(X)$, after using the manuscript's topological input $H^2(X, \mathbf{Z}) = 0$ and the asserted normalization $\widetilde{W} \simeq \mathrm{dP}_6$.
5. The base-change isomorphism transfers this nonvanishing to
$$R^2 f_*(T_X \otimes L) \neq 0.$$

No counterexample occurs in the properness-flatness-duality-base-change layer. A counterexample to the actual candidate must enter earlier: in the construction of the compact Hausdorff threefold, the toric description and normalization of W, or the integral topology used to obtain $H^2(X, \mathbf{Z}) = 0$.

0.2 Status discipline

The supplied workbench record reports exact-check harnesses but also records two proof-layer transcription corrections and expressly keeps compressed toric and nearby-cycle steps open until they survive line-by-line reconstruction. This update preserves that discipline.

The statuses used here are:

- **verified:** a complete argument is given in this report;
- **verified-exact:** a finite algebraic identity also has a machine-reproducible certificate;
- **verified-from-input:** the deduction is complete after the named candidate inputs are supplied;
- **open candidate-specific:** the claim concerns the actual construction and is not replaced by rank or matrix agreement.

0.3 Properness, surjectivity, and flatness

Let X be a compact connected complex manifold of dimension three and let $f : X \to B = \mathbf{P}^1$ be a nonconstant holomorphic map.

0.3.1 3.1 Properness and surjectivity

A continuous map from a compact space to a Hausdorff space is proper. Thus f is proper.

A nonconstant holomorphic map from a connected complex manifold to a Riemann surface is open. Its image is also compact, hence closed in $\mathbf{P}^1$. The nonempty image is both open and closed, so $f(X) = \mathbf{P}^1$.

0.3.2 3.2 Flatness of f

Fix $x \in X$, set $b = f(x)$, and write
$$A = \mathcal{O}_{B,b}, \qquad R = \mathcal{O}_{X,x}.$$
The ring A is a discrete valuation ring. The ring R is a regular local integral domain because X is a complex manifold.

Let $t \in A$ be a local parameter. Its pullback $f^*t \in R$ is nonzero. If its germ were zero, f would map a neighbourhood of x into the point b; a holomorphic map locally constant on a nonempty open subset of connected X is globally constant, contrary to the hypothesis.

Since R is a domain, f^*t is a non-zero-divisor. Every nonzero element of the DVR A is a unit times a power of t, so no nonzero element of A annihilates a nonzero element of R. Thus R is torsion-free over A, and torsion-free modules over a DVR are flat. Hence f is flat at every point.

This proof includes a multiple fibre locally written $t = s^m$: the scheme-theoretic fibre is nonreduced, but s^m remains a non-zero-divisor.

0.3.3 3.3 Flatness of vector bundles

For a holomorphic vector bundle E, one has locally $E_x \simeq R^{\oplus r}$. Since R is flat over A, E_x is flat over A. Therefore every vector bundle on X, including $T_X \otimes L$, is f-flat.

The morphism proved flat is exactly

$$\mathcal{O}_{B,b} \longrightarrow \mathcal{O}_{X,x};$$

the proof identifies the image of a uniformizer as a non-zero-divisor. It does not assert that f is smooth.

0.4 Top-degree cohomology and base change

Every scheme-theoretic fibre X_b has pure complex dimension two: locally it is cut out in the regular three-dimensional local ring R by the non-zero-divisor f^*t.

Work on a sufficiently small Stein neighbourhood $U \subset B$ of b. The analytic cohomology-and-base-change theorem represents $Rf_*E|_U$ by a bounded complex $K^\bullet$ of finite free $\mathcal{O}_U$-modules, and f-flatness identifies its derived fibre with ordinary coherent cohomology of the scheme-theoretic fibre:

$$K^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}_U} k(b) \simeq R\Gamma(X_b, E|_{X_b}).$$

For a bounded complex of finite free modules over $A = \mathcal{O}_{B,b}$, the universal-coefficient exact sequence is

$$0 \longrightarrow H^q(K) \otimes_A k(b) \longrightarrow H^q(K \otimes_A k(b)) \longrightarrow \operatorname{Tor}_1^A(H^{q+1}(K), k(b)) \longrightarrow 0. \tag{4.1}$$

0.4.1 4.1 Vanishing of the next direct image

Suppose $H^q(K) \neq 0$ for some $q > 2$, and choose the largest such q. At a point in its support, Nakayama gives $H^q(K) \otimes k(b) \neq 0$. The maximality of q makes the Tor term in (4.1) zero, so the fibre has nonzero H^q. A coherent sheaf on the compact two-dimensional complex space X_b has no cohomology above degree two. This contradiction gives

$$R^q f_* E = 0 \qquad (q > 2).$$

0.4.2 4.2 The degree-two isomorphism

For $q = 2$, the obstruction in (4.1) is

$$\operatorname{Tor}_1^A(R^3 f_* E, k(b)) = 0.$$

Therefore

$$(R^2 f_* E)_b \otimes_A k(b) \xrightarrow{\sim} H^2(X_b, E|_{X_b})$$

for every b. No constancy of h^2, local freeness of $R^2 f_* E$, reducedness of the fibre, or smoothness of f is used.

This identifies the exact reason top degree works: the only possible defect is the Tor term from the next direct image, and that next direct image vanishes.

0.5 Duality on the central fibre

Now use the candidate-specific fibre $W = f^{-1}(p_0)$, asserted to be a reduced effective Cartier divisor of multiplicity one in the smooth threefold X.

A Cartier divisor in a smooth space is a local complete intersection, hence Cohen-Macaulay and Gorenstein. Adjunction gives

$$\omega_W \simeq (\omega_X \otimes \mathcal{O}_X(W))|_W.$$

Since $W = f^*(p_0)$ as a Cartier divisor,

$$\mathcal{O}_X(W) \simeq f^*\mathcal{O}_B(p_0).$$

Its restriction to W is the pullback of the one-dimensional vector space $\mathcal{O}_B(p_0)|_{p_0}$ along the constant map $W \to p_0$, so it is a trivial line bundle. Hence

$$\omega_W \simeq \omega_X|_W.$$

For

$$G = (T_X \otimes L)|_W, \qquad A = (L^* \otimes \omega_X)|_W,$$

Serre-Grothendieck duality on the compact Cohen-Macaulay surface gives the exact isomorphism

$$H^2(W, G)^\vee \simeq H^0(W, G^\vee \otimes \omega_W) \simeq H^0(W, \Omega^1_X|_W \otimes A). \tag{5.1}$$

Thus it remains to construct one nonzero section on the right.

0.6 The conductor-supported differential

Let $\eta : \widetilde{W} \to W$ be the normalization, $D \subset W$ the reduced double locus, and $\widetilde{D} = \eta^{-1}(D)$. The candidate supplies the local models

$$t = x, \qquad t = xy, \qquad t = xyz.$$

0.6.1 6.1 Conormal injection

Since W is Cartier, the conormal sequence is exact:

$$0 \longrightarrow \mathcal{O}_W \cdot dt \longrightarrow \Omega^1_X|_W \longrightarrow \Omega^1_W \longrightarrow 0. \tag{6.1}$$

The included finite checks certify injectivity directly in the singular models:

$$((xy) : x) \cap ((xy) : y) = (y) \cap (x) = (xy),$$

and

$$((xyz) : yz) \cap ((xyz) : xz) \cap ((xyz) : xy) = (x) \cap (y) \cap (z) = (xyz).$$

0.6.2 6.2 Vanishing on the conductor and nonvanishing off it

In the double model,

$$d(xy) = y\,dx + x\,dy,$$

whose coefficients lie in the conductor ideal (x, y). In the triple model,

$$d(xyz) = yz\,dx + xz\,dy + xy\,dz,$$

whose coefficients lie in (xy, xz, yz). Therefore $\eta^*(dt)|_{\widetilde{D}} = 0$.

Away from D, the model is $t = ux$ with u a unit, so $dt|_W = u\,dx \neq 0$.

0.6.3 6.3 Normalization-conductor descent

The normalization-conductor sequence is

$$0 \longrightarrow \mathcal{O}_W \longrightarrow \eta_*\mathcal{O}_{\widetilde{W}} \oplus \mathcal{O}_D \longrightarrow \eta_*\mathcal{O}_{\widetilde{D}} \longrightarrow 0. \tag{6.2}$$

The double and triple inclusion-exclusion reconstructions are reproduced in the exact-check script.

Assume $\eta^*A \simeq \mathcal{O}_{\widetilde{W}}$, and choose a nowhere-zero trivializing section e. Tensoring (6.2) with $E = \Omega^1_X|_W \otimes A$, the pair

$$(\eta^*(dt) \otimes e, 0)$$

agrees on $\widetilde{D}$ and descends uniquely to

$$s \in H^0(W, \Omega^1_X|_W \otimes A).$$

It is nonzero because its pullback is nonzero off $\widetilde{D}$. Its image in $\Omega^1_W \otimes A$ vanishes on the dense smooth locus and is conductor-supported torsion. This is the exact nonnormal-fibre mechanism.

0.7 Triviality on the normalization

The manuscript uses

$$H^2(X, \mathbf{Z}) = 0.$$

Therefore $c_1(M) = 0$ for every line bundle M on X. With the candidate-specific identification $\widetilde{W} \simeq \mathrm{dP}_6$, one has $H^1(\widetilde{W}, \mathcal{O}) = 0$. The exponential sequence makes

$$c_1 : \mathrm{Pic}(\widetilde{W}) \hookrightarrow H^2(\widetilde{W}, \mathbf{Z})$$

injective. Naturality gives

$$\eta^*(M|_W) \simeq \mathcal{O}_{\widetilde{W}}$$

for every $M \in \mathrm{Pic}(X)$, in particular for $M = L^* \otimes \omega_X$. Thus the descent construction applies to the line bundle A in (5.1).

Combining duality and descent gives

$$H^2(W, (T_X \otimes L)|_W) \neq 0. \tag{7.1}$$

Combining (7.1) with top-degree base change gives

$$R^2 f_*(T_X \otimes L) \neq 0. \tag{7.2}$$

0.8 Exact dependency boundary

The verified arrows begin at three named inputs:

CAND-X: compact connected smooth X and nonconstant $f : X \to \mathbf{P}^1$

$\Downarrow$

properness, flatness, and vector-bundle f-flatness

$\Downarrow$

β_b^2 is an isomorphism for every scheme-theoretic fibre,

and

$$
\begin{array}{c}
\text{CAND-W: reduced Cartier SNC fibre, } \widetilde{W} \simeq \mathrm{dP}_6 \\
\text{TOP-H2: } H^2(X, \mathbf{Z}) = 0 \\
\Downarrow \\
\eta^*((L^* \otimes \omega_X)|_W) \simeq \mathcal{O}_{\widetilde{W}} \\
\Downarrow \\
\text{conductor-supported section} \\
\Downarrow \\
H^2(W, (T_X \otimes L)|_W) \neq 0 \\
\Downarrow \\
R^2 f_*(T_X \otimes L) \neq 0.
\end{array}
$$

The arrows after the three inputs have been checked. The inputs are not established by these arrows, and Section 10 must not be used circularly as evidence for TOP-H2.

0.9 Preliminary audit of the candidate construction

A targeted rereading of Theorem 4.5 and Theorem 6.2 found no immediate defect in the following chain:

- $B_0 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ is integral, orthogonal, and unimodular;
- fixed-point exclusion uses an explicit norm inequality and a bounded holomorphic correction;
- proper discontinuity reduces chart intersections to a bounded lattice-translation problem;
- the quotient is a covering and Hausdorff by finite-translate separation;
- a finite union of compact toric polydiscs supplies a compact fundamental set over each smaller closed disc;
- the local fibre equations descend through a local biholomorphism as x, xy, and xyz;
- the global gluing uses pairwise disjoint punctured base discs, so nonsingleton equivalence classes have exactly two representatives and the transitions are the displayed biholomorphisms.

This is a positive audit finding, not a complete certification of Sections 3-6. Every analytic input defining the period functions and the two finite-point fillings still needs the same line-by-line treatment.

0.10 Candidate-specific test program

0.10.1 CAND-X: existence of the global threefold

Target: the quotient and gluing produce a second-countable compact Hausdorff smooth complex threefold and a global holomorphic map.

Decisive falsifiers: a nonfree point; infinitely many deck translates meeting a fixed compact set; a nonsingle-valued overlap; a branch change altering the glued complex structure; a non-Hausdorff pair; or failure of local properness at a special point.

0.10.2 CAND-W: exact central fibre

Target: the quotient fibre is reduced SNC, has the stated three double curves and two triple points, and has normalization dP_6 with the printed conductor identifications.

Decisive falsifiers: a ray of nonunit height; a nonprimitive quotient identification; an extra orbit of vertices, edges, or triangles; hidden isotropy inside a branch; a wrong pairing of boundary curves; or a quotient chart not biholomorphic to x, xy, or xyz.

0.10.3 TOP-H2: integral topology

Target: an integral calculation of $H^2(X, \mathbf{Z})$ in which each matrix is proved to be the matrix of the actual geometric morphism in the stated bases.

Decisive falsifiers: a sign or nonprimitive meridian in based van Kampen; a different drift; a nonsaturated image; a transgression not equal to the displayed integer p; or a mismatch between a printed matrix and the geometric map.

0.10.4 ANA-REF: publication-level references

Target: bind the finite-complex theorem, analytic Serre-Grothendieck duality, and the nonnormal conductor formalism to exact cited statements with every hypothesis matched.

This is a source-binding task, not a newly found mathematical gap in the reconstructed arrows.

0.11 Conclusion

The properness-flatness-duality-top-base-change chain is valid. The chain starts only after the candidate's global construction, central-fibre description, and integral H^2-vanishing have been supplied.

Base change is therefore no longer a plausible candidate-specific weak point. The highest-value unresolved work is the construction and integral topology, especially the exact geometric identification of the nearby-cycle/Leray and based van Kampen maps.

0.12 Package contents

The evidence package contains this report, a claim-by-claim ledger, a candidate-specific test-program ledger, exact symbolic checks for the xy and xyz models, source excerpts for the relevant manuscript sections, the supplied current-progress transcript, and a fail-closed verification script.

The exact script certifies only the finite identities it names. It does not certify the global candidate inputs.